

HAT-P-18b

R-2200

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_210518 · created 2026-08-14T00:44:56+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459352.78291 +/- 0.00027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.08 +/- 0.05
Transit depth (Rp/Rs)^2	0.63 +/- 0.79 [%]
Orbital Inclination (inc)	87.86 +/- 1.84
Airmass coefficient 1 (a1)	1.034 +/- 0.023
Airmass coefficient 2 (a2)	-0.025 +/- 0.039
Scatter in the residuals of the lightcurve fit is	2.26 %
Best Comparison Star	#2 - [142, 150]
Optimal Aperture	3.98
Optimal Annulus	5.5
Transit Duration (day)	0.076 +/- 0.041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.08 ± 0.05 vs 0.1356 (deviation 1.11σ)
Duration fit vs published	0.076 d vs 0.1129 d (deviation 0.9σ)
Mid-time O–C	-0.9 min
Depth SNR	1.6
Residual scatter	2.26 %

Light curve

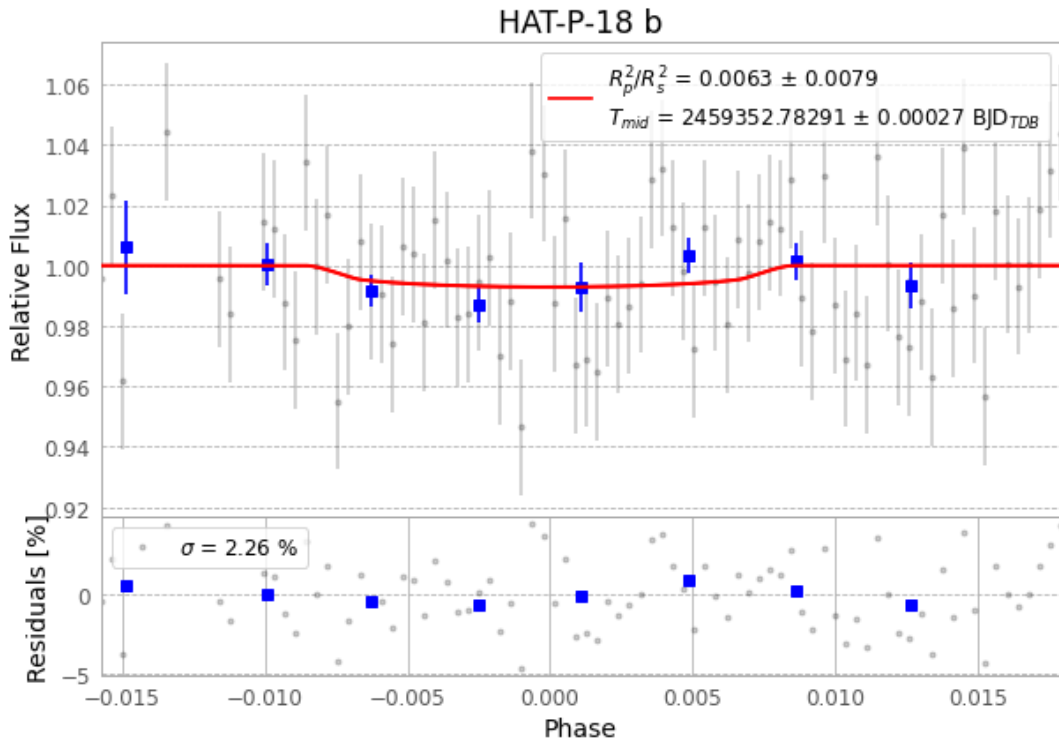


Figure 1 — FinalLightCurve_HAT-P-18 b_2021-05-17.png (SHA-256 6023055f1d5be803...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [334, 229], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a35b571b4d8cb9f5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

90 FITS frames (59 MB), HATP-18210518043612.FITS → HATP-18210518090312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-210518.log (c3cde2e778be...), FinalLightCurve_HAT-P-18 b_2021-05-17.png (6023055f1d5b...), FinalLightCurve_HAT-P-18 b_2021-05-17.csv (186838884b26...)

Compute resources

Wall time 403.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 00:44Z.